Mass-Conserving Multi-Task AI for Crop Yield and Nitrogen Outcomes

Team briefing | CAPRI-first research plan | 9 October 2026

Objective

Test whether one model that learns across multiple crops and respects a nitrogen budget can predict crop yield, harvested nitrogen removal, and nitrogen surplus more accurately and consistently than models without that constraint.

Why this is a plausible research gap

Prior work already predicts maize yield and N loss with machine learning [1], and DeepCGM uses mass conservation for crop-growth variables while leaving the nitrogen cycle for future work [2]. The proposed contribution is the combined, cross-crop nitrogen-budget test using a common European dataset. This is a targeted novelty claim, not proof that no equivalent study exists; repeat the literature search before finalising a paper.

How the model works

Each crop-area-year record enters a shared encoder (crop identity, location, year, weather, soil, management and N inputs). Three connected outputs estimate yield, harvested N removal and surplus. Crop-specific embeddings or output heads allow different crop responses while sharing information across crops. Train against the recorded targets and a nitrogen-budget consistency term.

> N inputs = harvested N removal + other terms subtracted by the chosen CAPRI budget + N surplus

Choose one CAPRI budget boundary (for example, soil-surface surplus) and follow its exact components and units [3]. Surplus is the budget remainder and can include unmeasured losses or soil N stock change; it is not itself a measured pollution flux. A hard equality is suitable only when every required term is observed or modelled. Otherwise use a weighted penalty and report residuals; do not claim strict conservation.

Inputs, outputs and first scope

Inputs	Outputs
CAPRI crop, area, location, year; mineral/manure N, biological fixation, deposition and other budget terms; optional soil and weather context.	Yield (t/ha); harvested N removal (kg N/ha); soil-surface N surplus (kg N/ha/year). Keep all N terms on one area and time basis.

Start with EU + UK, 2000-2018, and a small set of well-covered crops (for example, wheat, barley, grain maize and rapeseed). Work at CAPRI Farm Structure Unit or aggregate to NUTS2 if spatial allocation noise is too high. Audit crop-by-year coverage and whether surplus can be assigned consistently to a crop before training [3].

Comparison and evaluation

Compare (1) independent crop-specific regressors, (2) a shared three-output model without a constraint, and (3) the same model with a budget penalty or constrained output. If feasible, add a simple budget-calculation baseline. Report MAE/RMSE and bias for each target; N-budget residual and violation rate; crop and country breakdowns; and accuracy versus constraint strength. Use blocked tests: later years (for example 2017-2018) and held-out regions; test unseen crops separately only if enough examples remain. Tune on earlier years only and avoid spatial neighbours leaking between train and test.

Data, impact and practical limits

Primary source first; supporting datasets are for context or external checks.

Exact data and source links

Primary: CAPRI disaggregated dataset - Crop areas/yields and N inputs, outputs and surplus components; EU-wide, 2000-2018; CC BY 4.0.
https://zenodo.org/records/13992301

GlobalCropYield5min - Independent yield comparison for wheat, maize, rice and soybean; 1982-2015, 5 arcmin.
https://doi.org/10.17632/hq8wzgx4yp.3

European N-surplus (Batool et al.) - Regional or country-level external context; 1850-2019. Different land coverage/budget definitions require careful aggregation.
https://zenodo.org/records/6581441

SoilGrids - Soil texture, organic carbon, pH and total N context; static maps. Use WCS or WebDAV while the REST API is paused.
https://docs.isric.org/globaldata/soilgrids/index.html

ERA5-Land - Temperature, precipitation and land-surface weather covariates; aggregate hourly data to crop-relevant seasons.
https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land

Optional: HaNi-Crop - Crop-specific fertiliser/manure N input comparison; 0.1 degree annual grids, 1961-2023. Check licence and harmonisation.
https://zenodo.org/records/21364978

AI-for-Good value

The model could highlight crop-region-years with high yield and lower N surplus, helping research on more efficient nitrogen use and potential water and climate benefits. Surplus is an indicator of potential loss, not a direct measurement of nitrate leaching or N2O emissions. Any fertiliser recommendation would need independent field validation and an intervention study.

Key limitations to record up front

CAPRI values are partly modelled and spatially disaggregated, so a high score against CAPRI may show reproduction of CAPRI assumptions rather than field truth [3]. Yield, N removal and surplus are algebraically linked; if the model sees the exact components used to calculate a target, apparent accuracy can be tautological. Avoid target leakage, identify the CAPRI surplus definition, harmonise kg N/ha/year and crop-area denominators, and compare external datasets only after matching their spatial scale and budget boundary. SoilGrids is largely static and cannot explain year-to-year change. GlobalCropYield5min ends in 2015 [4].

References

[1] Shahhosseini et al. (2019), maize yield and nitrate loss ML. doi:10.1088/1748-9326/ab5268
[2] Han et al. (2025), DeepCGM mass-conserving crop growth. article
[3] Koeble et al. (2026), CAPRI dataset documentation and N-budget definitions. doi:10.1038/s41597-026-06919-8
[4] Cao et al. (2025), GlobalCropYield5min description. doi:10.1038/s41597-025-04650-4
[5] Batool et al. (2022), European soil N-surplus dataset. doi:10.1038/s41597-022-01693-9